SCALE100X.AI · PRODUCT TAKE-HOME

Task 1: Product Thinking

Franchise-in-a-Box OS

At a glance: the questions, the answers, the numbers

1a Prioritisation: what ships in the first 30 days

1b AI vs human split: is "76% AI" real?

1c Developer-ready spec: Conversation Takeover

Appendix: assumptions and decisions

15 October 2026

At a glance

What Task 1 asked

Part	The question, in plain words
1a	If only part of the platform can be built in the first 30 days, what goes first, and why?
1b	The brief says the platform runs on "76% AI, 24% human". Is that realistic?
1c	Take one feature the brief describes vaguely and write it up so a developer could build it without guessing.

The answers

1a: Build the part that earns revenue first. That means an AI WhatsApp sales agent that answers new franchise leads within minutes, plus a simple CRM where the sales team takes over and closes. The ROI calculator comes next, then brand onboarding. Why: Scale100x is paid when a franchisee signs, and the pilot brands are already onboarded, so the sales path can start on day 1. A checkpoint on day 5, using real sales data, confirms the order or changes it.

1b: Believable for workload, misleading as worded. Mapping all 37 tasks on the platform shows automation covers 68–87% of the work, so 76% is plausible. But only about 54% is genuinely AI; about 34% is ordinary software (templates, schedules, formulas). And people still make every high-value decision. The recommendation: replace the fixed label with a measured, audited figure: **"AI: X% of work | Humans: 100% of closes"**.

1c: A developer-ready spec for "human takeover" of AI chats. The brief gives this one line. The spec defines who owns a conversation at every moment and what happens when two people grab it or nobody does (a 10-minute response rule with escalation). It also covers WhatsApp's 24-hour messaging rule and opt-out, with 10 edge cases and 8 pass/fail tests.

Headline numbers

Measure (per brand, per month, modelled)	Without AI	With AI
Hours of work	457	59
Brands one consultant can handle	0.35	2.7
Human hours per signed franchisee	~152	~20
Share of work time automated	—	87% (54% AI + 34% software)

These come from a model with stated estimates (lead volumes, minutes per task). A pilot checks them; the method is the point, not the exact percentages.

How the work was done

- Modules were ranked on five weighted criteria, each score with a written reason.
- The 37-task map lives in a live Google Sheet; every number in 1b comes from its formulas.
- WhatsApp platform rules were checked against Meta's own documentation.
- Every assumption and decision is labelled (A- and D- numbers) and listed in the appendix.
- The three parts fit together: the 1c spec is the core of the 1a plan, and it records the events used to measure 1b.

Terms used

Term	Meaning
Brand / franchisee	The brand sells franchises; a franchisee buys one and opens an outlet.
Lead	A prospective franchisee who has shown interest, usually through a Facebook or Instagram ad form ("Lead Ads").
CRM	The system the sales team uses to track leads and deals.
HITL	Human-in-the-loop: a person steps into a conversation the AI is handling.
ROI	Return on investment: what a franchisee can expect to earn back, and how fast.
SLA	A committed response time (here, a flag must be claimed within 10 business minutes).
BSP	A Meta-approved provider of WhatsApp Business API access.
Autonomy Rate	The proposed audited measure of how much work AI completes with no human edit.

Reading the rest: each part opens with a TL;DR box, then goes into the detail product and engineering reviewers need. **Online versions:** live Sheet · Google Doc · GitHub

1a. Prioritisation: what ships in the first 30 days

> **TL;DR.** First, ship the lead-to-close slice: the WhatsApp Sales Agent and a thin CRM, released together. Next comes a thin ROI Engine, so the agent has approved numbers to quote, then thin Onboarding. Collateral Studio waits. The order follows the money: Scale100x is paid when a franchisee signs (A1), and its 20+ existing brands (A21) mean the revenue path can start on day 1. The ranking is a hypothesis, not a fixed bet. Days 1–5 test it with discovery while the team does only work that every outcome needs, and a decision gate on day 5 says what changes if the evidence points elsewhere.

1. The lens

Scale100x's stated goal is to help brands open **100+ offline outlets a year**, and every one of those outlets is a signed franchisee. Scale100x earns a setup fee and retainer from brands, and a success fee when a franchisee signs (A1). The modules split into two groups:

- **Brand-side modules (Onboarding, ROI, Collateral)** earn the right to sell.
- **Franchisee-side modules (Sales Agent, CRM)** produce the signed agreements. They are also where most of the AI leverage sits: about 73% of all manual work per brand is in the Sales Agent module (1b).

Every module is ranked by one question: *does it get more qualified franchisees to a signed agreement, faster, across many brands at once?* A module doesn't rank higher just because it contains more AI.

2. Which bottleneck? Four hypotheses

The brief describes five modules but doesn't say which customer problem hurts most. This ranking rests on hypothesis H1, and discovery tests all four:

#	Hypothesis	What it would look like in the data	Module it would promote
H1	**Lead handling:** leads get slow replies or no follow-up	Long time to first reply; leads with no second touch; conversion falls with reply delay	Sales Agent + CRM (current #1)
H2	**Lead quality:** ads attract people who can't or won't invest	Many replies, few qualify; low-budget or wrong-city leads dominate	Qualification rules and lead-source analysis inside the Sales Agent
H3	**Investor confidence:** prospects stall at the numbers	Drop-off after the investment or payback conversation; repeated ROI questions	ROI Engine moves to #1 alongside the agent
H4	**Brand readiness:** brands lack the facts and documents to sell	Consultants can't answer basic questions; missing P&L or FAQs	Onboarding moves above ROI

H1 is the working bet, for two reasons:

- The money is earned at signing, and leads go cold within hours, not days.
- The 1b action map puts most of the manual work in the conversation itself.

That is a model of the work, not observed pain. It still has to be confirmed with funnel data.

3. How the modules are scored

Criterion	Weight	Question
Business impact	30%	Does it move franchisees closer to signing?
Customer pain	25%	Does it fix a frequent, costly problem?
30-day feasibility	20%	Can a useful, testable version ship in 30 days with this team (A7)?

Criterion	Weight	Question
Dependency leverage	15%	Do other modules need it?
Evidence confidence	10%	How much evidence backs the need today?

These weights are proposed decision weights, not measured results. The scores below (1–5) are pre-discovery judgments. Each comes with a written reason so it can be challenged, and all of them are re-scored after day 5.

Module (thin scope)	Impact	Pain	Feasibility	Leverage	Evidence	**Weighted**
HITL Sales CRM	4	4	4	4	3	**3.90**
AI Sales & Marketing Agent	5	4	3	3	3	**3.85**
Franchise Model & ROI Engine	3	3	4	4	2	**3.25**
Brand Onboarding	2	2	5	4	2	**2.90**
Collateral Studio	2	2	3	1	2	**2.05**

Reasons for the scores that matter most:

- **Sales Agent impact 5:** it is the step where a lead becomes a conversation.
- **Sales Agent feasibility 3:** Meta approvals are outside the team's control (§6).
- **CRM leverage 4:** the agent's takeover and audit both depend on its timeline.
- **ROI leverage 4:** its approved Model Card is the only source of numbers the agent may quote (A32).
- **Onboarding impact 2:** the pilot brands are already onboarded (A21).
- **Collateral leverage 1:** no other module needs it to work.

The CRM and the Agent score within 0.05 of each other, which is why they share rank 1 and ship as one slice.

4. Ranking (decision D-04, confirmed)

Team: 2 full-stack engineers, 1 designer and a PM, using AI tools on React + Supabase (A7). Pilot: 2 existing brands (A21).

Rank	Module	30-day scope (what ships)	Why now	Left out until later
1	AI Sales & Marketing Agent (thin)	WhatsApp AI Agent only: Lead Ads webhook → dedupe by phone number → consent → template greeting → qualification → answers from approved facts → HITL flag and takeover; basic live tracking (counts of leads, messages and conversations per brand)	Leads stop going cold because the reply arrives in minutes; this is where 73% of the work is	Voice and Email agents, Meta Ads optimisation, the full switchboard, live-tracking dashboards beyond basic counts
1	HITL Sales CRM (thin), shipped as one slice with the agent	Inbox, owner per lead, unified timeline, next action with follow-up reminders, takeover panel, deal stage up to Won / Lost	Without a human closing path the agent produces chats, not signed agreements	Live call intelligence, forecasting
3	ROI Engine (thin)	Franchise Model Card + 3-year ROI with editable, labelled assumptions, approved by a consultant and published as the agent's **approved facts**	Prospects' first serious question is about returns. The agent may only quote numbers a human has approved (A32).	Benchmarks (shown only when sourced, A33), AI insights
4	Brand Onboarding (thin)	Brand details form, file upload, required-field checks; a consultant enters data for the 2 pilot brands	Needed to add brand #3 onwards, but the pilot brands are already onboarded	AI Readiness Score, revenue-split donut

Rank	Module	30-day scope (what ships)	Why now	Left out until later
5	Collateral Studio	Deferred: the pilot uses each brand's existing deck plus a small set of approved WhatsApp templates	Existing brands already have decks; designers can make creatives with current tools	Everything

Why ROI ranks above Onboarding. A reasonable case can be made for the opposite order, because the agent needs approved brand information to answer well (H4). The two modules split that information differently:

- **Onboarding** holds brand details (name, category, cities). For the 2 pilot brands, a consultant can type these in.
- **ROI** holds the information prospects actually press on: investment, fee, royalty, payback. That is the part that has to be right and approved.

So ROI is the module that unblocks good answers. If discovery finds the pilot brands' basic facts are missing or inconsistent (H4), the two swap places.

Reuse before build: scheduling uses a calendar-booking link, and reminders use the CRM's own task list. The team doesn't build email or calendar tools in the first 30 days.

5. The dependency trap

The brief lists the modules in brand-journey order: Onboarding → ROI → Collateral → Sales → CRM. Building in that order looks logical, but it means the **first lead isn't handled until three brand-side modules have shipped**. At our capacity (A7) that falls outside the 30-day window, and the month would end with polished onboarding screens and no signed franchisees.

Reversing the order works because the pilot brands are **already onboarded** (A21). Their data can be entered by hand while the slice that earns the success fee goes live.

6. Day-1 critical path: WhatsApp access

The slice depends on outside approvals that the team can't speed up:

- WhatsApp Business API access through a BSP
- Meta business verification
- approval of the greeting and re-engagement **message templates**

Templates matter because a free-form message can only be sent within 24 hours of the prospect's last message. Outside that window, Meta returns error 131047 and only a template can be sent (Meta error codes).

These applications go in on day 1, in parallel with discovery and the build. No lead time is quoted here because none has been verified. The fallback if approval is late: run the pilot on the CRM with consultants replying by hand, and switch on the agent once access lands. The CRM work isn't wasted either way.

7. The 30-day roadmap

Days	Work	Exit gate
1–5: Discover, and start the no-regret work	**Discovery:** interview the pilot brands' sales consultants; pull whatever funnel data exists (lead volume, time to first reply, lead → call → signing); map the current lead lifecycle and where leads drop. **No-regret work** (needed under all four hypotheses): submit the BSP and template applications; build the CRM data model and the `ACTION_EVENT` log (1b §5); enter the pilot brands' data.	**Decision gate (day 5):** which hypothesis does the evidence support? Re-score §3, confirm or change the ranking, and record baselines for the §8 metrics.
6–12: Workflow foundation	Lead intake from Lead Ads, dedupe, owner and status, CRM timeline, follow-up reminders, access control, error handling	A test lead flows end to end with a human replying

Days	Work	Exit gate
13–21: AI assistance and handoffs	Agent qualification and answers grounded in approved facts; low-confidence and off-topic handling; flag → takeover (the 1c state machine); thin ROI Model Card published as approved facts	The 1c acceptance tests AC1–AC8 pass on a scripted test set
22–26: Pilot	Go live on the 2 pilot brands; run the first weekly audit (1b); fix defects; thin Onboarding form in parallel (it isn't on the pilot's critical path)	0 wrong answers in the audit; flags are being claimed
27–30: Measure and decide	Compare against the day-5 baselines; review the audit, adoption and time saved	Decide what comes next: add a channel, deepen the CRM, or promote ROI

What changes at the day-5 gate:

- **H1 confirmed:** keep the plan.
- **H2 (lead quality):** keep the slice, but move effort from answering to qualifying, with stricter qualification rules and reporting by lead source. A faster reply to the wrong lead doesn't help.
- **H3 (investor confidence):** pull the ROI Model Card into days 6–12 and make it the agent's first answer about returns.
- **H4 (brand readiness):** swap Onboarding and ROI, and make required brand facts a go-live check.

8. Day-30 success criteria

These are targets for the 2 pilot brands, measured from the event log. They are product goals, not observed data. If the day-5 gate produces a real baseline, each target is re-set against it.

#	Criterion	Target
1	Pilot brands live on the slice	2
2	Median time to first **meaningful** reply on a new lead	≤ 5 minutes
3	Leads that reach a qualification outcome (qualified / not qualified / not now)	≥ 70%
4	Discovery calls booked through the agent	≥ 15 across both brands
5	Flags claimed within the 10-business-minute SLA (D-06)	≥ 90%
6	Agent answers that contradict the approved facts (weekly audit, 1b)	0

A **meaningful reply** answers the lead's question or asks a qualification question. A plain acknowledgement ("Thanks, we'll be in touch") doesn't count toward criterion 2, even if it's instant.

Also tracked from day 1, without targets until there's a baseline:

- follow-ups completed on time
- duplicate lead records
- missed escalations (from the audit)
- consultant hours per lead, which tests the 1b capacity model

9. How the team uses AI to deliver this

The team uses AI to build faster, in three parallel workstreams that share one data model and integrate daily:

- **Workstream 1:** inbox and CRM UI.
- **Workstream 2:** webhook, data and agent.
- **Workstream 3:** the test set, which is a script of realistic WhatsApp conversations for each pilot brand, used to score the agent's answers.

The aim is a working end-to-end slice inside the first week. Speed of writing code isn't the constraint, though. The things that take real time are Meta approvals, permissions, data quality and the human handoff. Those are proven by the pilot and the audit, not by how fast the code was written.

10. Risks and mitigations

Risk	Mitigation
H1 is wrong: lead handling isn't the main bottleneck	Day-5 gate with pre-agreed switches (§7); days 1–5 do only work that every hypothesis needs
Meta approvals are slow	Apply on day 1; run the pilot on the CRM with human replies until access lands (§6)
The agent gives a wrong answer about money	The agent quotes only approved facts (A32); uncertain or off-script questions get flagged to a human; weekly audit with a 0% threshold
Flags aren't picked up in time	10-business-minute SLA with escalation (1c); criterion 5 tracks it
Pilot brands' lead data is hard to get	Use a scripted test set for the build, and real leads only once the webhook is live
Consultants don't adopt the inbox	Involve them in discovery; the takeover panel replaces their WhatsApp phone, so they don't enter data twice
ROI numbers read as a promise	Assumptions are labelled and editable; benchmarks are shown only when sourced (A33); no guaranteed returns
AI-written code ships defects	Code review, automated tests on the takeover state machine, and the 1c acceptance tests as the release gate
Scope grows past 30 days	Everything outside the §4 thin scope goes to the "later" column; nothing joins it without something else leaving

1b. AI vs Human split: is 76% real?

> **TL;DR.** We mapped all 37 actions across the five modules. Depending on how you count, automation does **68% to 87%** of the work, so "76%" is plausible as a measure of workload. Three things make it misleading as stated:
>
> - **Not all of it is AI.** Of the 87% of time that's automated, AI models do 54% and ordinary software rules (templates, schedules, formulas, logging) do 34%.
> - **It ignores value.** Humans own every one of the highest-value decisions.
> - **It isn't stable.** It swings 14 points depending on how often humans edit the AI's drafts.
>
> The action map makes 76% plausible but doesn't prove it. Our fix is an audited **Autonomy Rate**, computed from logged events and validated in a pilot against a baseline. It's shown as "AI: X% of work | Humans: 100% of closes" (D-05), with the model-vs-rules split on hover, and always paired with outcome metrics.

1. Action map

All 37 user-facing actions are listed in `ai-human-split.csv`, which imports straight into the Sheet tab `AI_Human_Split`. `ai-human-split.xlsx` is the same data with every formula below already built in, and it's live as a Google Sheet. Each action has an owner:

- **AI:** done end to end by AI.
- **AI→H:** AI drafts, a human approves.
- **H:** human only.

Owner says who is accountable. A second column, **engine**, says what kind of automation does the work, so ordinary software isn't counted as AI:

- **model:** an AI model does it (extraction, drafting, conversation, summaries).
- **rules:** fixed software does it (templates, schedules, formulas, logging, rule-based scores).
- **human:** no automation.

Module	Actions	AI	AI→H	H	Manual hours / brand / month (no AI)	Human hours left with AI
1 Onboarding	7	4	1	2	1.2	0.3
2 ROI Engine	7	4	1	2	2.8	1.0
3 Collateral Studio	6	5	1	0	16.0	1.0
4 Sales & Marketing Agent	10	9	1	0	334.7	2.0
5 HITL CRM	7	3	1	3	102.0	54.6
Total	**37**	**25**	**5**	**7**	**456.6**	**58.8**

Human-only actions, and why they stay human:

- **Onboarding:** uploading the files; the go/no-go decision to take a brand on.
- **ROI Engine:** agreeing the fee, royalty and investment with the brand; presenting the model to the founder.
- **CRM:** taking over a flagged chat; running the discovery call; negotiating and signing.

All four actions with the top value weight (5) are human.

Who does what, by module (from the engine column):

Module	AI models	Software rules	Humans
1 Onboarding	Pre-fill the brand form (founder confirms); extract P&L figures; explain gaps and fixes	Revenue-split chart; Readiness Score v0 (a rule-based rubric, A31)	Upload files; go/no-go on the brand
2 ROI Engine	Draft the Model Card (consultant approves); find comparable benchmarks	3-year projections and sensitivity (formulas); insights v0 (rule-based, per the prototype C plan)	Agree fee, royalty and investment; present to the founder
3 Collateral	Generate deck, brochure, ads and WhatsApp scripts; regenerate; draft the send (human approves)	—	Approve before anything goes out
4 Sales Agent	Qualify on WhatsApp; answer FAQs from approved facts; voice qualification; detect flags; draft ad changes (human approves)	Capture, consent and dedupe; template greeting; email sequence; follow-up nudges; live counts	Approve ad spend changes
5 CRM	Takeover summary; live call notes; suggest deal stage (human confirms)	Interaction timeline	Take over chats; discovery call; negotiate and sign

Counts: 14 actions are AI-model only, 11 are software rules, 5 are AI drafts with human approval (all model-drafted), and 7 are human only.

Inputs are estimates, not data.

- Monthly volumes come from A19: 500 leads, about 150 voice calls, 60 takeovers, 20 discovery calls and 3 signings per brand.
- Standard minutes and the 1–5 value weights come from A20.
- One-off brand set-up actions are spread over a 6-month engagement (A34).

All of these get calibrated in the pilot. The value of this section is the method, not the precise percentages.

2. The split, measured three ways

Method	Formula (Sheet tab `AI_Human_Split`, rows 2–38)	Result
Count, strict (AI only)	`=COUNTIF(D2:D38,"AI")/COUNTA(D2:D38)`	**68%** (25 / 37)
Count, generous (AI→H credited to AI)	`=(COUNTIF(D2:D38,"AI")+COUNTIF(D2:D38,"AI→H"))/COUNTA(D2:D38)`	**81%** (30 / 37)
Time-weighted (share of manual minutes done by AI)	`=SUMPRODUCT(E2:E38,F2:F38,G2:G38)/SUMPRODUCT(E2:E38,F2:F38)`	**87%**
Value-weighted (share of value weight done by AI)	`=SUMPRODUCT(H2:H38,G2:G38)/SUM(H2:H38)`	**68%**
Time-weighted, **AI models only**	`=SUMPRODUCT((J2:J38="model")*E2:E38*F2:F38*G2:G38)/SUMPRODUCT(E2:E38,F2:F38)`	**53.6%**
Time-weighted, **software rules only**	same, with `"rules"`	**33.5%**

Columns: D = owner, E = frequency per month, F = manual minutes, G = AI share of the minutes, H = value weight, J = engine.

By value, the 68% splits into 45.7% models and 22.4% rules. The rules share is mostly high-volume Sales Agent plumbing: the email sequence alone is 50 of the 153 hours that rules take over.

How sensitive the time-weighted number is

Change to the inputs	Time-weighted AI share	Human hours / brand / month
Base case	87.1%	58.8
Agent conversations take half as long to do by hand	80.0%	57.8
Leads halved to 250 a month	79.4%	58.8
Takeovers doubled to 120 a month	83.8%	78.8
Every AI→H approval saves only 50%	86.6%	61.4
Humans edit 20% of the agent's work	**72.8%**	**124.1**
Day-30 slice only (D-04): unshipped AI work done by people	**76.5%**	**89.6**

Two findings matter most:

- **The human edit rate moves the number more than any volume assumption.** That's why the Autonomy Rate below only credits AI for work no human had to edit.
- **The day-30 slice lands at about 76%.** This assumes the agent's work isn't edited and that consultants and designers still do the onboarding, collateral and ad work by hand. The brief's figure is reachable early, but only as a measured outcome, not a constant.

3. Verdict: is 76% AI realistic?

As a claim about…	Verdict	Why
Workload	**Yes**	68–87% depending on method; 76.5% for the day-30 slice
Value	**No, not as stated**	Value-weighted share is 68%. Humans own every top-value decision: go/no-go, terms, discovery call, close.
A live number at launch	**Only if measured**	It's 76.5% at day 30 and 87% at full build, so it changes as modules ship
"AI" specifically	**No, not as worded**	AI models do 54% of the time; 34% is ordinary software that any CRM could do. A badge that says "AI" for all of it overstates the AI.
A hardcoded label	**No**	A 20% edit rate moves it 14 points. A fixed number would be wrong most of the time, and an investor can't audit it.

Testing the brief's own claims

The brief says…	What the action map shows	Holds?
"AI-driven automation (70–80%)"	68–87% depending on the method; 76.5% for the day-30 slice. AI models alone: 54% of time.	**Yes for automation; no for "AI-driven".** The range holds only if templates, schedules and formulas count as AI.
"Human intervention only for high-value actions"	6 of the 7 human-only actions have value weight 4–5. The exception is uploading the deck and P&L (row 2, weight 1), which only the founder can do because only they hold the files. Humans also approve 5 AI drafts, and 2 of those are low-value (brand details form, deal stage, both weight 2).	**Almost.** The upload can shrink (pull files from a shared drive) but not disappear. The 2 low-value approvals should become AI-only once the audit shows the drafts are accurate.
"AI: 76% | Human: 24%"	Human share is 13% of the time, 32% of the value and 32% of the actions (strict count), so "24%" doesn't match any single measure. And 93% of the remaining human hours (54.6 of 58.8) are in the CRM: takeovers, discovery calls and closing.	**No.** "Human: 24%" presents people as overhead to squeeze, when their hours are where the revenue event happens. That's why the badge becomes "Humans: 100% of closes" (D-05).

4. What this means commercially: consultant capacity

Without AI, one brand needs about **457 hours of work a month**. With AI, about **59 human hours** remain, almost all of them in CRM: takeovers, discovery calls and closing. At 160 productive hours per consultant per month (A35):

- **Without AI:** about **0.35 brands per consultant**, roughly 3 people per brand.
- **With AI:** about **2.7 brands per consultant**, about 8× the leverage.
- **If humans edit 20% of the agent's work:** about **1.3 brands per consultant**.

This is the investor story, not the percentage. Leverage is measured in brands per consultant and hours per signed franchisee: about **20 human hours** per signing with AI versus about **152** without (59 h and 457 h ÷ 3 signings).

5. The fix: an audited Autonomy Rate

Definition (A5)

Autonomy Rate = the share of standard minutes of funnel work completed by AI **with no human edit**, per brand, over a rolling 28 days.

```
Autonomy Rate = Σ std_minutes(AI actions, not edited) ÷ Σ std_minutes(all actions)
```

Three numbers, kept apart

The word "automation" covers three different measures. They're never mixed in one figure:

Measure	Formula	Answers	Used for
A. Autonomy Rate (the badge)	AI-completed std minutes, unedited ÷ all std minutes in scope	How much of the whole workload runs without people?	The badge X
B. Success rate on eligible work	AI completions that pass the audit ÷ AI attempts	When AI is allowed to act, how often is it right?	Deciding which actions may run unapproved
C. Human time saved	(baseline human minutes − human minutes now) ÷ baseline	Did people actually get time back?	The capacity story in §4

In this model, A and C come out equal (87%) because the human minutes are derived from the same table. In the pilot they will differ: reviews, rework and corrections add human minutes that A doesn't see. That's why C is measured against a real baseline, not computed from the table.

Unedited-approval rule

- An **AI** action counts as AI.
- An **AI→H** action counts as AI only if the human approves it **unchanged**.
- If the human edits it, rejects it or rewrites it, it counts as human.
- A conversation that gets flagged counts the takeover minutes as human.
- An AI action that fails the weekly audit (wrong answer, missed escalation) counts as human, because someone had to fix it.

Event schema

Every action writes one `ACTION_EVENT`, which matches the platform data model:

Field	Type	Notes
`id`	uuid	
`brand_id`	uuid	Rate is computed per brand, then rolled up
`action_type`	enum (37 values)	Matches `ai-human-split.csv`

Field	Type	Notes
`actor`	`ai_model` | `software` | `human`	Who completed it. Splitting AI models from software keeps the badge honest.
`ai_draft_edited`	bool	True if a human changed an AI draft before it went out
`approved_by`	uuid, nullable	For AI→H actions
`std_minutes`	int	Looked up from a **versioned** standard-minutes table, never typed in by hand
`std_minutes_version`	string	So a past rate can be recomputed after a recalibration
`conversation_id`	uuid, nullable	Links to the audit sample
`ts`	timestamptz	

Audit

- **Weekly sample:** a lead consultant reviews **50 randomly sampled AI-handled conversations** across the live brands against a fixed rubric.
- **Two error rates are published next to the badge:**
  - **Missed-escalation rate:** a conversation should have been flagged and wasn't.
  - **Wrong-answer rate:** the agent said something that isn't in the approved facts.
- **Thresholds:** if missed escalations go above 5% or wrong answers above 0% in a week, the badge shows "under review" until the prompt or rules are fixed. These are proposed product thresholds, to be set with the founder.
- **Recalibration:** standard minutes are recalibrated each quarter from timed samples in the pilot (A20). Each change gets a new version.

Investor view

The badge keeps the brief's format but changes what it means (D-05):

> **AI: X% of work | Humans: 100% of closes**

X is the live Autonomy Rate. Hovering over it shows the definition, the split between AI models and software rules, the audit date and the two error rates. Next to it go three outcome metrics, so autonomy is never shown alone:

1. **Human hours per signed franchisee:** modelled at about 20 h, versus about 152 h without AI.
2. **Cost per signed franchisee:** ad spend + AI cost + human hours × loaded hourly cost, divided by signings. The inputs come from the pilot; no number is claimed here.
3. **Speed to lead:** median minutes to first meaningful reply (not a plain acknowledgement). The target is ≤ 5 (1a).

6. Validating 76%: the pilot plan

76% is treated as a hypothesis to test, not a capability to claim. The pilot is the 2 brands in the day-30 slice (1a §7).

1. **Baseline first (days 1–5).** For each in-scope action: how many instances a month, how long a person takes now, error and rework rates, and what the brand's existing tools already automate. Existing automation is counted as the baseline, never as new AI.
2. **Log every instance.** Every action writes an `ACTION_EVENT`: attempted, completed, escalated, failed, edited.
3. **Count only successes.** An AI action counts only if it's unedited and passes the audit (see the rule above). A fast reply with a wrong fact is a failure, not automation.

4. **Report with the fine print.** Each published rate states the period, the brands, which actions were in scope, what was excluded, and the audit results. Model and rules shares are always shown separately.
5. **Expand by evidence.** An action moves from AI→H to AI only when its success rate (measure B) stays above the agreed bar for 4 consecutive weekly audits. The bar is a proposal to agree with the founder.

Guardrails during the pilot:

- wrong money or commercial claims (target 0)
- missed escalations (≤ 5%)
- duplicate leads
- messages sent without consent

Any breach pauses the badge ("under review").

1c. Developer-ready spec: Conversation Takeover (HITL)

> **TL;DR.** The brief describes human takeover in one line: "Includes Human-in-the-Loop (HITL) takeover". That one line hides the hardest behaviour in the product: who owns a live WhatsApp conversation, what happens when two people grab it, and what happens when nobody does. This spec defines 7 states and 9 transitions, a 10-business-minute claim SLA (D-06), the WhatsApp 24-hour window, opt-out, 10 edge cases, 8 testable acceptance criteria, and the events it writes for the Autonomy Rate (1b). Prototype B (Task 2) implements it.

How we chose the feature and built the spec

Step 1: List the features the brief describes in a line or less. Each one hides at least one open question that a developer would have to guess at.

Step 2: Screen them with three questions.

- Does it ship in the day-30 slice (1a)?
- What does it cost if a developer guesses wrong?
- Is the unclear part *what to measure* or *how it behaves*?

A measurement can be shipped as a v0 and tuned later. A behaviour has to be right on day one.

Feature (as the brief puts it)	What's unclear	In the day-30 slice?	Cost of a wrong guess	Unclear part
HITL takeover ("Includes Human-in-the-Loop (HITL) takeover")	Who owns the chat; what happens when two people grab it; what happens when nobody does; the 24-hour window; opt-out	**Yes**, rank 1	**High**: a lost hot lead, or a double reply from AI and a human	**Behaviour**
AI Readiness Score (RAG sub-scores)	What is scored, the weights, the thresholds	No (score deferred)	Medium: a rubric can be iterated (rule-based v0 in prototype A, A31)	Measurement
AI insights (Opportunity / Risk / Benchmark)	What triggers an insight	No (deferred)	Medium: misleading advice; v0 is rule-based	Measurement
Live call intelligence	What it captures and suggests, and when	No (deferred)	Medium	Behaviour
Meta ads optimisation	What the AI may change without approval	No (deferred)	High (ad spend), but outside the 30-day scope	Behaviour
Live tracking	Which numbers to show	Yes (basic counts)	Low: counts can be added later	Measurement

Step 3: Pick the feature that is in the slice, expensive to get wrong, and unclear in behaviour. Only takeover meets all three. The final call was between takeover and the Readiness Score (A18): *takeover is harder to build; the Readiness Score is harder to define.* The 1b action map confirms the choice. Takeover is where human work begins: 93% of the human hours left after AI (54.6 of 58.8 a month per brand) are in the CRM, starting with takeovers (1b).

Step 4: Write it in the order a developer needs it.

1. Users and stories
2. States
3. Transitions, each with a trigger, a guard and a system action
4. UI by state
5. Error, empty and loading states

6. Validation
7. Edge cases
8. Acceptance tests with numbers
9. Metrics after launch

Step 5: Check platform facts against the source. The 24-hour window, error 131047 and the 4,096-character limit were checked against Meta's documentation (see the end of this spec). The SLA and business hours are product choices, labelled as such (D-06, A22).

Step 6: Check consistency with 1a and 1b.

- The holding message isn't a "meaningful reply" (1a §8).
- The spec covers the AI-off fallback from 1a §6.
- Each step writes an `ACTION_EVENT` with the 1b actor split (§10).

Step 7: Hand over to prototype B. Prototype B (Task 2) builds the state machine, and its test script runs AC1, AC2, AC4, AC5 and AC7.

1. Users and stories

As a...	I want to...	So that...
Consultant	see which conversations need me, why, and how long I have	I reach high-intent prospects before they go cold
Consultant	claim a conversation and know nobody else is replying	the prospect never gets two answers
Consultant	hand a conversation back to the AI with a reason	the AI keeps nurturing, and we learn why the flag fired
Sales manager	be alerted when a flag sits unclaimed past the SLA	no hot lead is dropped
Prospect	get a human reply quickly when I ask for one, and stop messages when I say STOP	I trust the brand

v1 scope: WhatsApp only, a single brand per conversation, one consultant owns a conversation at a time, and English / Hindi / Hinglish keyword lists.

Not in v1: voice and email takeover, round-robin auto-assignment, consultant skill routing.

2. States

State	Who may send messages	Meaning
`AI_ACTIVE`	AI only	Default. The AI qualifies and answers from approved facts.
`FLAGGED`	Nobody (the AI has sent one holding message)	A HITL rule fired. Waiting for a consultant to claim it. The SLA clock is running.
`ESCALATED`	Nobody	Unclaimed past the SLA. The manager is alerted.
`HUMAN_ACTIVE`	The assigned consultant only	The AI is paused for this conversation.
`CLOSED_WON`	Nobody (terminal)	Deal won.
`CLOSED_LOST`	Nobody (terminal)	Deal lost.
`OPTED_OUT`	Nobody, including templates (terminal)	The prospect sent STOP.

3. State transitions

Every transition increments `conversation.state_version` and writes a `STATE_EVENT` row (who, from, to, reason, ts).

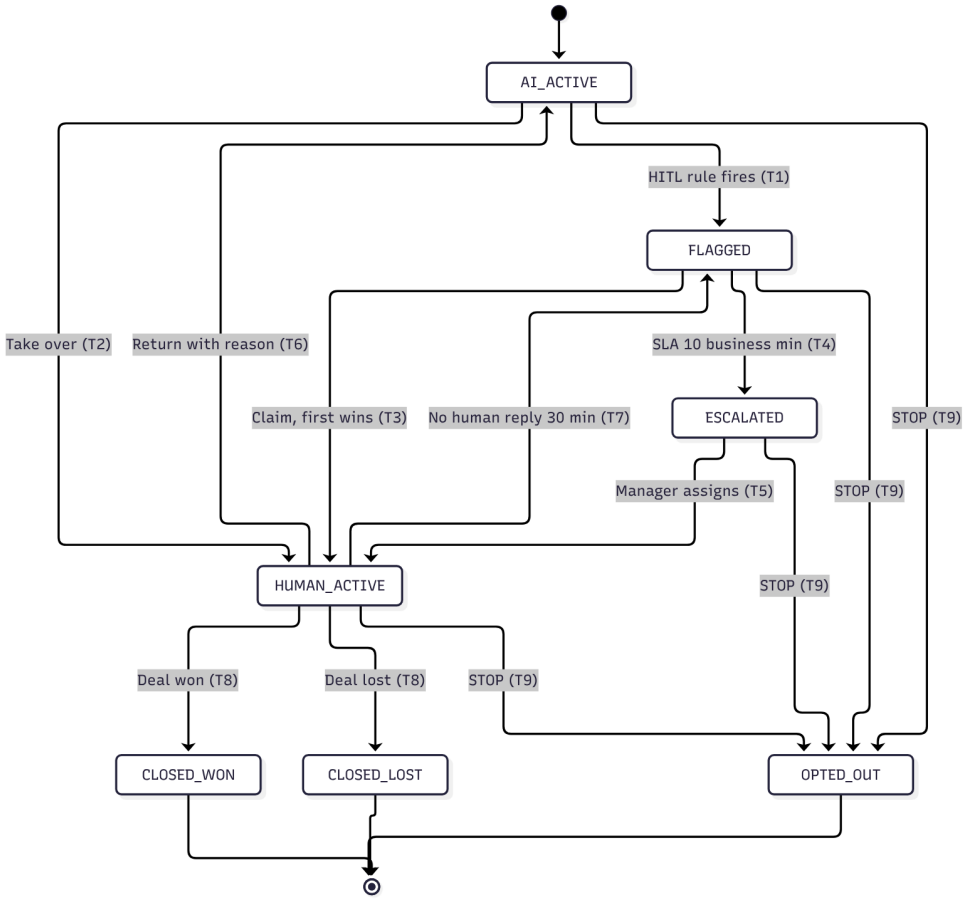
#	From → To	Trigger	Guard	System action
T1	AI_ACTIVE → FLAGGED	An inbound message fires ≥ 1 HITL rule (Task 3 triggers)	Not opted out	Create `FLAG` with every fired `rule_id` and the values that fired; set `sla_due_at` = now + 10 business min; AI sends **one** holding message; notify on-duty consultants
T2	AI_ACTIVE → HUMAN_ACTIVE	Consultant clicks **Take over** without a flag	User role is consultant or manager; `state_version` matches	Assign to the user; AI paused; "AI paused" system note in the thread
T3	FLAGGED → HUMAN_ACTIVE	Consultant clicks **Claim**	`state_version` matches (first claim wins)	Assign; stop the SLA clock; record claim time
T4	FLAGGED → ESCALATED	`sla_due_at` passes (business minutes only)	Still unclaimed	Notify the brand's manager (in-app + email); conversation moves to the top of the manager's queue
T5	ESCALATED → HUMAN_ACTIVE	Manager assigns a consultant, or claims it	`state_version` matches	Assign; record the escalation duration
T6	HUMAN_ACTIVE → AI_ACTIVE	Consultant clicks **Return to AI**	A reason is selected; every compliance flag (legal / financial) is marked **Resolved**; the composer is empty	AI resumes on the next inbound message; the reason is stored on the flag
T7	HUMAN_ACTIVE → FLAGGED	Prospect's last message is unanswered for 30 business minutes	Last message is from the prospect	New flag, reason "No human reply"; new SLA clock; the original assignee keeps first claim for 2 minutes, then anyone can claim
T8	HUMAN_ACTIVE → CLOSED_WON / CLOSED_LOST	Consultant sets the deal stage to Won or Lost	Lost requires a loss reason	Close the deal record; lock the composer
T9	any non-terminal → OPTED_OUT	Inbound "STOP", "unsubscribe", "band karo", "message mat karo" (case-insensitive, exact-message match)	—	Send one confirmation (the only permitted message); write `CONSENT_RECORD` = withdrawn; block all sends, including templates; cancel the SLA clock

Business minutes: 09:00–21:00 IST, every day in v1. This is configurable per brand and is a product choice (A22, D-06). A flag raised outside those hours starts its clock at 09:00, and the holding message tells the prospect when to expect a reply.

Holding message (EN; HI and Hinglish versions are also stored):

> "Thanks, a member of our franchise team will reply to you here shortly." Outside hours, "shortly" becomes "after 9 am".

The holding message is an acknowledgement, not a **meaningful reply**, so it doesn't count toward the speed-to-lead target in 1a §8. For a flagged conversation, the clock stops at the consultant's first reply.



State diagram: the 7 states and transitions T1–T9 from the table above.

4. UI behavior

Inbox (left column)

- Each row shows: prospect name or number, brand tag, last message preview, and a state chip:
  - AI: blue `#2563EB`
  - Flagged / Escalated: amber `#F59E0B`
  - Human: green `#10B981`
  - Closed / Opted out: grey
- The **Needs human** filter shows FLAGGED and ESCALATED conversations, sorted by SLA time remaining, earliest due first.
- The SLA countdown is shown in mono type and turns red under 2 minutes.
- New flags reach every open session within 2 seconds through realtime.

Conversation header

- Prospect name, brand, state chip, assignee avatar, and the 24-hour window countdown (`window_expires_at`).
- Buttons by state:

State	Buttons shown
AI_ACTIVE	**Take over**
FLAGGED	**Claim**
ESCALATED	**Claim** (managers also see **Assign**)
HUMAN_ACTIVE	**Return to AI**, **Mark Won**, **Mark Lost**

State	Buttons shown
Terminal states	None

"Why flagged" card (pinned at the top of the thread while FLAGGED / ESCALATED, collapsible afterwards)

- Lists each fired rule by name, with the exact value that fired. Example: *Deal value: stated investment ₹45L ≥ brand threshold ₹40L*; *Explicit request: "call me"*.

Takeover panel (right column, HUMAN_ACTIVE)

- A 5-bullet AI summary: budget, city, timeline, objections, last unanswered question.
- The lead score and one suggested reply. The suggested reply is inserted into the composer only on click and is never auto-sent.

Composer

State	Composer behaviour
AI_ACTIVE	Disabled. Placeholder: "AI is handling this conversation. Take over to reply."
HUMAN_ACTIVE	Enabled, with an "AI paused" banner and a character counter (`n / 4,096`)
Window closed	Free text is disabled and replaced by a template picker
Terminal states	Hidden. Read-only thread.

Thread

- Shows prospect, AI and human bubbles, plus system events inline: "Flagged: high intent", "Claimed by Priya", "Returned to AI: question answered".

5. Error handling: empty, loading and error states

Situation	What the user sees	System behaviour
Queue empty	"No conversations need you right now." plus the count currently handled by AI	—
Loading inbox or thread	Skeleton rows / bubbles; no spinners on the whole page	Thread loads its last 50 messages, then more on scroll
Send failure (BSP error or timeout > 10 s)	Bubble marked red "Not delivered", with **Retry**	Text stays in the bubble; Retry reuses the same `client_message_id`, so it can't send twice; at most 3 retries, then the conversation is flagged "Delivery problem"
Out-of-window send	Composer swaps to the template picker: "More than 24 h since the prospect's last message. Send an approved template."	If the API still returns error **131047**, the bubble shows "Window closed. Send a template." and free text is never retried automatically
Claim conflict	Toast: "Already claimed by Priya at 14:02". The view refreshes to their ownership.	Server rejects the stale `state_version` (HTTP 409); no state change
Realtime disconnect	Banner: "Reconnecting… messages may be out of date". **Send** is disabled until reconnected.	Resubscribes with backoff; on reconnect, refetches the state and messages since the last event id
Prospect messages while FLAGGED	The new message appears in the thread and the Why-flagged card updates if more rules fire	No second holding message; the SLA clock doesn't reset

6. Validation rules

Field / action	Rule
Message body	1–4,096 characters after trimming; whitespace-only is rejected. 4,096 is the WhatsApp text limit (Meta text messages).
Any state-changing action	Must send the current `state_version`; a mismatch returns 409
Return to AI	A reason is required: *Question answered* / *Not ready, keep nurturing* / *Wrong flag* / *Other* (if Other, free text of 5–200 characters). Every compliance flag must be Resolved first.
Mark Lost	A loss reason is required from a fixed list
Template send	Every template variable must be filled; only templates with status **approved** are listed
Any send	Blocked if `CONSENT_RECORD.status` = withdrawn, or the state is terminal
Claim / Take over	Role must be consultant or manager for that brand (row-level security)

7. Edge cases

1. **Two consultants claim at the same moment.** The first write wins through `state_version`. The second sees the claim-conflict toast.
2. **A flag fires at 20:58 IST.** About 2 business minutes count today, and the clock resumes at 09:00. The holding message says "after 9 am".
3. **Several rules fire on one message.** One flag lists every reason; there are never duplicate flags for the same message.
4. **A closed lead messages again.** A new conversation cycle opens in `AI_ACTIVE` with the full history kept, and the previous outcome is shown in the header.
5. **STOP during a human conversation.** T9 applies: the composer locks and the human sees "Prospect opted out". Only an inbound "START" can re-open the conversation, which writes a new consent record.
6. **The consultant goes offline mid-conversation.** If the prospect's message sits unanswered for 30 business minutes, T7 re-flags it.
7. **The window expires while the consultant is typing.** The draft is kept, Send switches to "Send as template", and nothing is sent as free text.
8. **The assignee is deactivated.** Their HUMAN_ACTIVE conversations re-flag immediately with reason "Assignee unavailable".
9. **The AI is switched off for a brand.** This happens before WhatsApp approval lands (the fallback in 1a §6), or when a manager pauses the agent. Every inbound message goes straight to FLAGGED through T1, with reason "AI off for brand". The same SLA applies, and no AI messages are sent apart from the holding message.
10. **The same prospect arrives as a new lead.** A Lead Ads submission whose phone number matches an open conversation for the same brand attaches to that conversation as a timeline event. It doesn't create a second conversation or reset the state.

8. Success criteria: acceptance tests

#	Given / when	Then (measurable)
AC1	AI_ACTIVE, an inbound message matches ≥ 1 rule	State is FLAGGED within **2 s** (p95). The card lists **every** fired rule with its value. Exactly **1** holding message is sent and **0** further AI messages follow.
AC2	Two claims on the same flag within **500 ms**	Exactly **1** succeeds. The other user sees the conflict toast within **1 s**. `state_version` increases by exactly **1**.
AC3	FLAGGED, unclaimed	Becomes ESCALATED at **10 business minutes** (± 30 s) and the manager is notified. A flag raised outside hours escalates at **09:10** at the earliest.

#	Given / when	Then (measurable)
AC4	HUMAN_ACTIVE, Return to AI with no reason or an unresolved compliance flag	The button stays disabled. **0** state changes.
AC5	Composer has **4,097** characters	Send is disabled and the counter is red. At **4,096** it sends.
AC6	> 24 h since the last inbound message	Free text is disabled and the template picker is shown. A **131047** response marks the bubble failed, with **0** automatic free-text retries.
AC7	"STOP" or "band karo" in any non-terminal state	OPTED_OUT within **2 s**. **0** messages are sent afterwards, templates included. A consent record is written.
AC8	HUMAN_ACTIVE, the prospect's message is unanswered	Re-flagged as "No human reply" at **30 business minutes** (± 30 s).

Prototype B's test script covers AC1, AC2, AC4, AC5 and AC7 directly. AC3 and AC8 use a shortened timer in the demo, and AC6 is simulated.

9. Success criteria: metrics after launch

Metric	Target (proposed, to be calibrated in the pilot)	Source
Flags claimed within the SLA	≥ 90% in month 1, ≥ 95% by month 3	`FLAG.sla_due_at` vs the claim `STATE_EVENT`
Median time to claim	≤ 5 business minutes	`STATE_EVENT`
Missed-escalation rate (weekly 50-conversation audit, 1b)	≤ 5%	Audit sheet
Double replies (AI and human within 60 s of each other)	0	`MESSAGE` log
"Wrong flag" share of Return-to-AI reasons	Tracked; above 30% triggers a review of the rule thresholds	Flag reasons
Median time from flag to the consultant's first reply (the holding message doesn't count)	≤ 10 business minutes	`MESSAGE` log
Takeovers that reach a booked discovery call	Baseline in month 1; no target until we have data	`DEAL` stage

10. Events this feature writes (for the Autonomy Rate)

Each step writes an `ACTION_EVENT` (1b §5). The table below shows how the steps count toward the badge.

Step	Action map row	`actor`	Counts as
Detect the rule and raise the flag (T1)	29 Detect HITL triggers	`ai_model`	AI. It becomes human if the weekly audit finds a missed escalation.
Send the holding message	23-style template	`software`	Automation, shown as software rules, not AI
Build the takeover summary	32 Summarise lead	`ai_model`	AI
Consultant replies, including a suggested reply sent unedited	33 Take over and reply	`human`	Human. A takeover is human work by definition (1b's unedited rule applies to AI→H drafts, not to takeovers).
Mark Won / Lost (T8)	37 Update deal stage	`human` (confirms an AI suggestion)	AI only if the suggested stage is accepted unchanged

Platform facts used (verified)

- **24-hour window and error 131047:** "More than 24 hours have passed since the recipient last replied to the sender number. Send the recipient a template message instead." (Meta error codes, checked 15 Oct 2026).
- **Text message limit:** 4,096 characters (Meta text messages; also stated in 360dialog's docs).

Appendix: assumptions and decisions

Every assumption (A) and decision (D) cited in this document. Assumptions are stated openly so they can be challenged; none of the inputs are presented as measured data.

Assumptions

ID	Topic	Assumption	Basis
A1	Business model	Scale100x earns a one-time setup fee, a monthly retainer, and a success fee per franchisee signed. Values are editable inputs.	Standard consultancy structure. Scale100x's pricing is **not public** (only job postings found).
A5	76% metric	Undefined in the brief. Computed three ways. Proposed **Autonomy Rate**: time-weighted share of funnel work completed by AI with no human edit.	Brief leaves it open
A7	Capacity	2 full-stack engineers + 1 designer + PM, using AI tools, building on React + Supabase from close to scratch	Brief gives no team size
A18	1c feature	Conversation Takeover chosen over the AI Readiness Score	Takeover is harder to build; Readiness is harder to define. Readiness is still prototyped in A (A31).
A19	Volumes	Per brand per month: 500 leads, about 150 voice calls, 60 takeovers, 20 discovery calls, 3 signed	Illustrative and internally consistent, **not sourced**
A20	Weights	Standard minutes and 1–5 value weights per action are candidate estimates, to be calibrated in the pilot	Estimate
A21	Pilot	30-day release goes to 2 existing brands	Scale100x already has 20+ onboarded brands
A22	SLA	Flag must be claimed within 10 business minutes; business hours 09:00–21:00 IST	Product choice, not research (D-06, confirmed)
A31	Readiness Score	Rule-based v0 rubric: 5 weighted sub-scores (unit economics, proof of concept, documentation, channel concentration, franchise terms), each Red/Amber/Green with the rule and threshold shown. Thresholds are design choices, to be calibrated on real outcomes.	The brief doesn't define the score
A32	Single source of numbers	Prototype C uses the Task 5 financial model's base-case inputs and formulas, so the same inputs give the same outputs. Prototype B's agent quotes only C's approved Model Card facts.	Avoids three different FreshBrew numbers
A33	Benchmarks	India + US benchmarks are shown only with a source and date; anything unsourced is labelled "illustrative" or left out	No benchmark data in the brief; no invented numbers
A34	Set-up amortisation	One-off brand set-up actions (onboarding, model, collateral) are spread over a 6-month engagement: frequency 1/6 per month in the action map	Estimate; makes per-brand-per-month hours comparable
A35	Consultant capacity	One consultant has 160 productive hours a month (8 h × 20 working days)	Estimate

Decisions

ID	Decision	Status	Date
D-04	30-day order: Sales Agent + CRM slice → thin ROI → thin Onboarding → Collateral deferred	Confirmed	15 Oct 2026
D-05	Badge reframed as "AI: X% of work | Humans: 100% of closes", where X is the live, audited Autonomy Rate (1b)	Confirmed	15 Oct 2026
D-06	Takeover SLA of 10 business minutes, 09:00–21:00 IST	Confirmed	15 Oct 2026